

# Supplementary Information

for

## Democratizing Reliable Knowledge-Seeking with MyGPT: A Privacy-First, Open-Source Retrieval-Augmented Generation Platform

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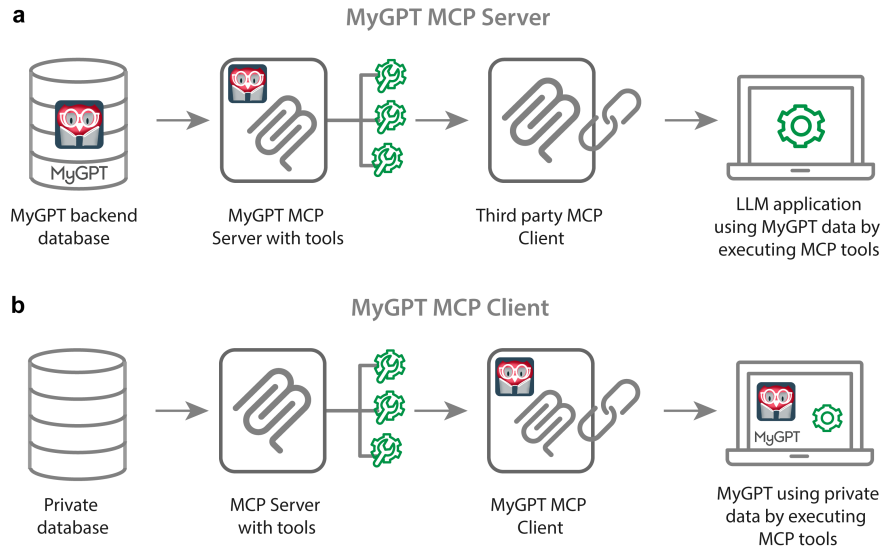
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### 1. Model Context Protocol

Model Context Protocol (MCP)<sup>1</sup> is an open standard developed by Anthropic that enables the integration of language model (LLM) applications with various external tools and data sources, both public and private. MCP operates on a client-server architecture. This means that information from external applications, local databases, or local files can be made available through an MCP server. In turn, LLMs from an AI application can access this information using the MCP client.

MyGPT provides a basic implementation of the MCP server and client, facilitating integration with other private LLM applications and data sources within the same AI ecosystem. The MyGPT MCP server exposes data from the MyGPT backend using MCP tools, which any third-party MCP client can utilize. Conversely, the MyGPT MCP client enables connections to the MCP server, whether it is running locally or in a private network (**Fig. 1**). This feature allows the client to use private database information as context, in addition to information from documents.



**Figure 1.** MCP server and client capabilities in MyGPT.

**a**, MCP server and **b**, MCP client workflow with MyGPT

The MyGPT server is built using a simple Python application that leverages the FastMCP<sup>2</sup> library, which is available as a Docker image for quick setup, similar to rest of the MyGPT pipeline. For prototyping purposes, we have developed two MCP tools: `get_mygpt_datasets` and `get_mygpt_documents`. To demonstrate how to use these MyGPT MCP tools, we employed a private third-party MCP client, such as the *MCP Client for Ollama* (*ollmcp*)<sup>3</sup>. As shown in **Figure 2**, users can inquire about the available libraries and the documents from MyGPT. The LLM, which can utilize these tools, can first execute the tool and then use the information generated to answer the user's questions.



**Figure 2.** MyGPT MCP server exposing information using MCP tools

The MCP server exposes data from the MyGPT backend to a third-party MCP client, such as *MCP Client for Ollama*, using MCP tools.

The MyGPT MCP Client allows users to connect to locally running or privately hosted MCP servers. Users can follow instructions on the MyGPT GitHub page to establish this connection. Once linked, the customization menu shows details about the MCP server and available tools. Users can select tools and compatible LLMs, like Llama3.1, Gpt-oss, Qwen3, etc., to prepare for real-time Q&A. For example, a demo MCP server was created using the MCP tools to utilize the Protein Data Bank (PDB). After connecting to this server and selecting Llama3.1:8b, users posed a complex question that required data from both the document and the Protein Data Bank. As shown in **Fig. 3**, MyGPT successfully collected the PDB ID from the document, used that information to retrieve additional data from the PDB database, and finally generated an answer, illustrating its capability to integrate various data sources for enhanced answers.



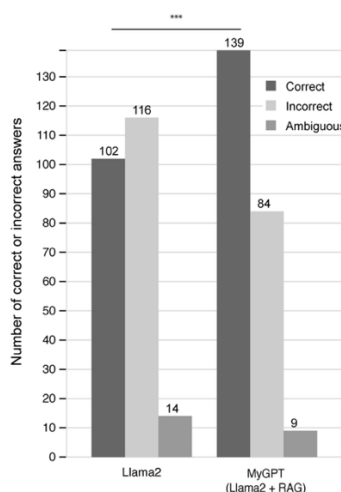
**Figure 3.** MyGPT MCP Client User Interface (UI) and Customization Menu

## 2. Benchmarking MyGPT on a dataset of questions and answers using public biomedical articles

To evaluate MyGPT we curated a dataset consisting of questions and correct answers (ground truths) from domain experts, along with a relevant set of publications to answer the questions. To do so, we asked 10 participants to provide us with at least 20 questions from 8-56 publications in their area of expertise. As a result, we collected 232 questions from 5 publication libraries (attached **Supplementary Material** of the entire dataset) covering different scientific domains within the area of drug discovery (G protein-coupled receptors (GPCRs), intrinsically disordered regions (IDRs), Chimeric Antigen Receptor – T Cells (CAR-T), Kinases, and Post-Translational Modifications (PTMs)). This evaluation was conducted from February to April 2024, during which Llama2:7b was the latest available open-source LLM. Out of 120 articles considered for the evaluation, 66 were published before the training cut-off date of 2022 for Llama2:7b, while 54 were published afterward, ensuring not all the questions can be answers with LLM alone and will need

RAG to work. This study design ensured that (a) not all answers to questions or information on the broad topic were expected to be found within the library (e.g., given the library size of documents is small); (b) the LLM models may have been partially trained on information in these documents; and (c) the LLM models alone would be unable to answer certain questions that rely on information from articles published after their training cut-off date. We do want to clarify that we did not use publicly available evaluation datasets, because of the possibility that those datasets are already included in the training of newer LLMs<sup>4</sup>.

The expert generated questions were answered by MyGPT using Llama2 with the relevant library and Llama2:7b (alone, without context). We used the all-MiniLM-L6-v2 embedding model and chunk size of 1000 characters without overlaps. We used the squared Euclidean distance metric to find the distance between the chunks and the question and Ollama's default setting for temperate (0.8), top-P (0.9) and top-K (40) to generate the answers. Participants were blinded to the response and were asked to provide feedback on whether the individual answers generated from MyGPT and Llama2 were correct, incorrect or ambiguous. We observed a 31.7% improvement in the number of correct answers generated by Llama2 when used within the MyGPT pipeline compared to Llama2 on its own for our dataset (**Fig. 4**).



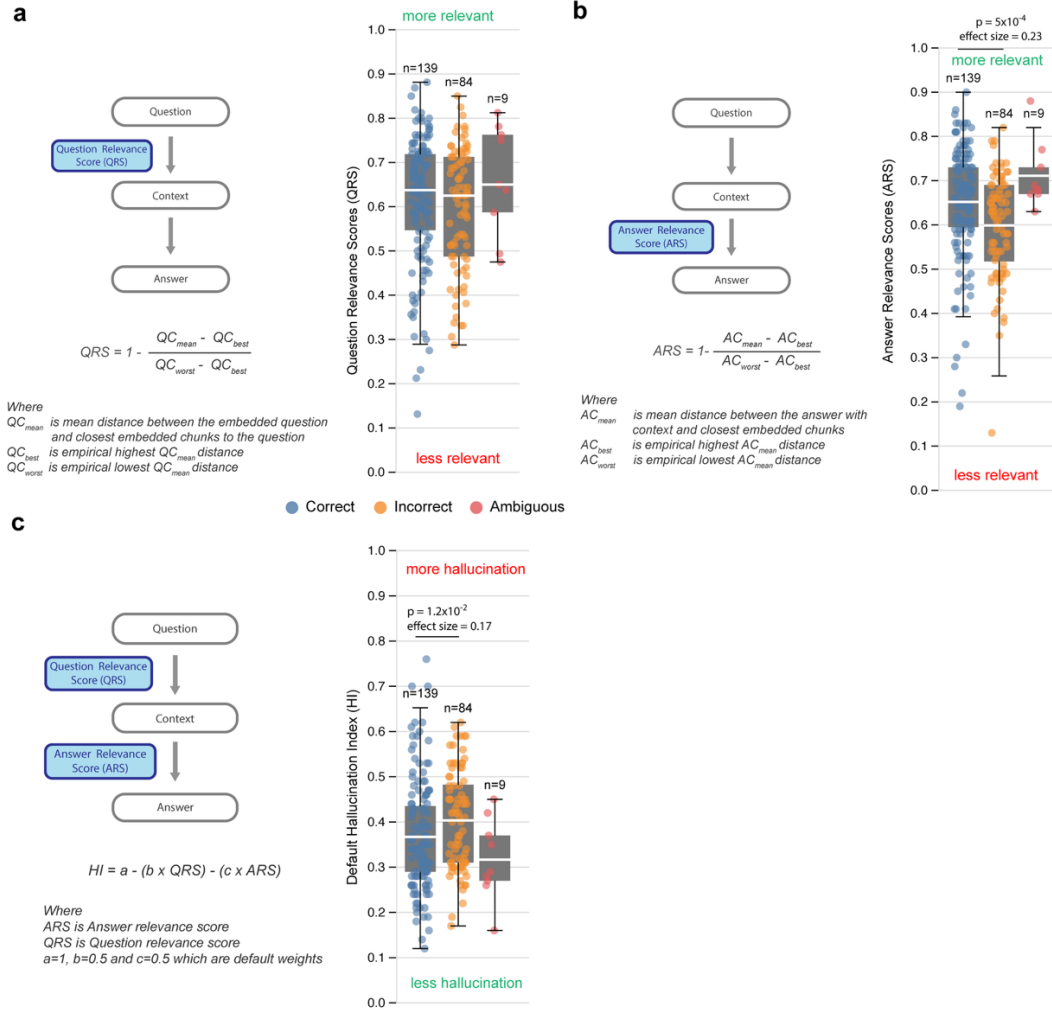
**Figure 4.** Performance of MyGPT generated answers on the dataset of public biomedical articles.

232 questions were collected from 10 domain experts from 5 publication libraries. Answers were generated using Llama2 (no context) and MyGPT (Llama2:7b + RAG). Frequency of answer correctness by standalone LLM and MyGPT. Users were blinded and asked if the generated answers were correct or not, and they provided feedback with “Yes” (dark grey), indicating the answer generated was correct, and “No” (light grey), indicating the answer generated was incorrect. Answers were considered “Ambiguous” if domain experts felt the generated answer was too vague to answer the intended question or if the answer was only partially correct.

We then analyzed the QRS, ARS, and the default HI distribution for correct and incorrect answers (**Fig. 5**). For our dataset of 232 questions from 5 publication libraries, we observed that the QRS scores were comparable for the questions with correct and incorrect answers. This suggested that for our dataset, the library of documents was generally relevant for the kinds of questions asked. In contrast, we observed that the ARS was higher for correct answers compared to incorrect answers ( $p = 5 \times 10^{-4}$ ), suggesting that correct responses tended to incorporate specific material from the library of documents while generating the response. Consistently, we also observed that the HI (default) was lower for correct answers compared to incorrect ones ( $p = 1.2 \times 10^{-2}$ ). The relationship between QRS, ARS, and HI for each question in the dataset shows how either higher QRS or ARS or both result in lower HI for correct answers compared to incorrect answers.

Taken together, these observations suggest that the QRS, ARS, and HI can serve as a quantitative guide to help identify whether the question is relevant to the material in the library, whether the answer has incorporated information from the library, and whether the response is likely to have been a hallucination. To enhance trustworthiness in AI-powered tools, users can manually evaluate their datasets, optimize the

HI scoring function, and use statistical learning approaches to leverage QRS, ARS, and HI scores to predict and classify responses as correct or incorrect. HI optimization and classification require generating a library, domain-specific questions, and labels indicating whether LLM-generated answers are correct or incorrect. By leveraging QRS, ARS, and HI, users can significantly improve the reliability and accuracy of answers generated by LLMs. We want to emphasize that correctness labels for LLM-generated answers are subjective. A user's preference for what constitutes a correct answer can result in varying ratios of correct to incorrect labels.



**Figure 5.** QRS, ARS and HI distribution of correct vs incorrect Q&A for biomedical dataset.

**a**, Question relevance score (QRS) distribution for correct and incorrect answers. **b**, Answer relevance score (ARS) distribution for correct and incorrect answers ( $p = 5 \times 10^{-4}$ ; effect size = 0.23, Wilcoxon rank-sum test). **c**, Hallucination Index for correct and incorrect answers using default parameters ( $p = 1.2 \times 10^{-2}$ ; effect size=0.17; Wilcoxon rank-sum test).

## References

- 1 Model Context Protocol (MCP) (GitHub <https://github.com/modelcontextprotocol>, 2024).
- 2 FastMCP (GitHub <https://github.com/jlowin/fastmcp>, 2024).
- 3 MCP Client for Ollama (ollmcp) (GitHub <https://github.com/jonigl/mcp-client-for-ollama>, 2025).
- 4 Cheng Xu, S. G., Derek Greene, M-Tahar Kechadi. Benchmark Data Contamination of Large Language Models: A Survey. *arXiv preprint arXiv:2406.04244* (2024). <https://doi.org/10.48550/arXiv.2406.04244>